

1. 证:

根据 Euler 方法:

$$u_1 = u_0 + h(a t_0 + b) \quad ①$$

$$u_2 = u_1 + h(a t_1 + b) \quad ②$$

$$u_3 = u_2 + h(a t_2 + b) \quad ③$$

$\vdots$

$$u_n = u_{n-1} + h(a t_{n-1} + b) \quad ④$$

①+②+...+④ 式得:

$$u_n = u_0 + h[a(t_0 + t_1 + \dots + t_{n-1}) + nb]$$

$u_0 = 0, t_i = hi, i = 0, 1, \dots, n-1$, 代入上式得

$$u_n = h[a(0 + 1 + \dots + (n-1))h + nb] = \frac{n(n-1)}{2} ah^2 + nbh$$

$$\therefore u(t_n) - u_n = \frac{n^2}{2} ah^2 + nbh - \left[\frac{n(n-1)}{2} ah^2 + nbh \right]$$

$$= \frac{1}{2} anh^2 = \frac{1}{2} aht_n$$

2. 解 =

$$\text{设 } f = u \cdot u, \text{ 则 } k_1 = u \cdot u_n, k_2 = u \left[u_n + \frac{h}{2} (u \cdot u_n + k_2) \right]$$

$$\text{故 } k_2 = \frac{(1 + \frac{uh}{2}) \cdot u}{1 - \frac{uh}{2}} \cdot u_n$$

$$\therefore u_{n+1} = u_n + \frac{h}{2} (k_1 + k_2)$$

$$= u_n + \frac{h}{2} \left[u \cdot u_n + \frac{(1 + \frac{uh}{2}) \cdot u}{1 - \frac{uh}{2}} \right] u_n$$

$$= \left[1 + \frac{uh}{2} + \frac{(1 + \frac{uh}{2}) uh}{2(1 - \frac{uh}{2})} \right] u_n$$

$$\text{令 } \bar{h} = uh, \text{ 则 } \lambda(\bar{h}) = 1 + \frac{\bar{h}}{2} + \frac{(1 + \frac{\bar{h}}{2}) \bar{h}}{(1 - \frac{\bar{h}}{2})^2} = \frac{1 + \frac{\bar{h}}{2}}{1 - \frac{\bar{h}}{2}}$$

$$|\lambda(\bar{h})| < 1 \Rightarrow -1 < \frac{1 + \frac{\bar{h}}{2}}{1 - \frac{\bar{h}}{2}} < 1$$

$$1) \bar{h} < 2 \text{ 时}, \frac{\bar{h}}{2} - 1 < 1 + \frac{\bar{h}}{2} < 1 - \frac{\bar{h}}{2} \Rightarrow \bar{h} < 0$$

$$2) \bar{h} > 2 \text{ 时}, 1 - \frac{\bar{h}}{2} < 1 + \frac{\bar{h}}{2} < \frac{\bar{h}}{2} - 1 \Rightarrow \text{无解}$$

综上, 绝对稳定区间为 $(-\infty, 0)$

3. 解:

$$(1) \text{ 原方程可代为 } \begin{cases} \frac{dX(t)}{dt} = AX(t) \\ X(0) = (2, 1)^T \end{cases}, \text{ 其中 } A = \begin{bmatrix} -0.1 & 1999 \\ 0 & -200 \end{bmatrix}$$

$$\det(\lambda I - A) = \begin{vmatrix} \lambda + 0.1 & -1999 \\ 0 & \lambda + 200 \end{vmatrix} = 0 \Rightarrow \lambda_1 = -0.1, \lambda_2 = -200$$

$\lambda = -0.1$ 时, 特征向量为 $(1, 0)^T$

$$\Rightarrow T = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$$

$\lambda = -200$ 时, 特征向量为 $(1, -1)^T$

$$\therefore X(t) = e^{At} X(0) = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} e^{-0.1t} & \\ & e^{-200t} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3e^{-0.1t} - e^{-200t} \\ e^{-200t} \end{bmatrix}$$

$$(2) S = \frac{1-2001}{1-0.11} = 2000$$

(3) 因为绝对稳定区间为 $(-2.78, 0)$

$$\bar{h}_1 = \lambda_1 h = -0.1h \quad \bar{h}_2 = \lambda_2 h = -200h$$

$$\text{故 } \begin{cases} -2.78 < -0.1h < 0 \Rightarrow 0 < h < 0.0139 \\ -2.78 < -200h < 0 \end{cases}$$

4. 解:

$$p(\lambda) = \lambda^2 + (b-1)\lambda - b = (\lambda+b)(\lambda-1) \quad \Delta(\lambda) = \frac{1}{4}(3+b)\lambda^2 + \frac{1}{4}(3b+1) \quad \text{令 } z = \lambda - 1$$

$$p(\lambda) - \ln(\lambda) \Delta(\lambda)$$

$$= z(z+1+b) - \ln(z+1) \left[\frac{1}{4}(3+b)(z+1)^2 + \frac{1}{4}(3b+1) \right]$$

$$= z^2 + (1+b)z - \left(z - \frac{1}{2}z^2 + \frac{1}{3}z^3 - \frac{1}{4}z^4 + \dots \right) \left[\frac{3+b}{4}z^2 + \frac{3+b}{2}z + (b+1) \right]$$

$$= z^2 + (1+b)z - \left[(1+b)z - \frac{1}{2}(1+b)z^2 + \left(\frac{3+b}{2}\right)z^2 + \left(\frac{3+b}{4}\right)z^3 - \frac{3+b}{4}z^3 + \frac{b+1}{3}z^3 - \frac{b+1}{4}z^4 + \frac{3+b}{6}z^4 - \frac{3+b}{8}z^4 + O(z^5) \right]$$

$$= -\frac{b+1}{3}z^3 + \frac{5b+3}{24}z^4 + O(z^5)$$

当 $b \neq -1$ 时, 为 2 阶, 截断误差为 $-\frac{b+1}{3}z^3 + O(z^4)$

当 $b = -1$ 时, 为 3 阶, 截断误差为 $-\frac{1}{12}z^4 + O(z^5)$

5. 解:

$$1) p(\lambda) = \lambda^2 - \frac{4}{3}\lambda + \frac{1}{3} \quad \delta(\lambda) = \frac{2}{3}\lambda^2$$

$$p(\lambda) - \bar{h}\delta(\lambda) = (1 - \frac{2}{3}\bar{h})\lambda^2 - \frac{4}{3}\lambda + \frac{1}{3} = 0 \quad \Leftrightarrow \lambda^2 - \frac{4}{3-2\bar{h}}\lambda + \frac{1}{3-2\bar{h}} = 0$$

$$|\lambda| < 1 \Leftrightarrow \left| \frac{4}{3-2\bar{h}} \right| < 1 + \frac{1}{3-2\bar{h}} = \frac{4-2\bar{h}}{3-2\bar{h}} < 2$$

$$\frac{4-2\bar{h}}{3-2\bar{h}} < 2 \text{ 成立, } \left| \frac{4}{3-2\bar{h}} \right| < \frac{4-2\bar{h}}{3-2\bar{h}} \Leftrightarrow 4-2\bar{h} > 4 \text{ 或 } 4-2\bar{h} < -4$$

可得 $\bar{h} < 0$, 则 $\bar{h} \in (-\infty, 0)$

$$2) p(\lambda) = \lambda^2 - \frac{4}{5}\lambda - \frac{1}{5} \quad \delta(\lambda) = \frac{2}{5}\lambda^2 + \frac{4}{5}\lambda$$

$$p(\lambda) - \bar{h}\delta(\lambda) = (1 - \frac{2}{5}\bar{h})\lambda^2 - \frac{4}{5}(1+\bar{h})\lambda - \frac{1}{5} = 0$$

$$\Leftrightarrow \lambda^2 - \frac{4(1+\bar{h})}{5-2\bar{h}}\lambda - \frac{1}{5-2\bar{h}} = 0$$

$$|\lambda| < 1 \Leftrightarrow \left| \frac{4+4\bar{h}}{5-2\bar{h}} \right| < 1 - \frac{1}{5-2\bar{h}} = \frac{4-2\bar{h}}{5-2\bar{h}} < 2$$

$$\frac{4-2\bar{h}}{5-2\bar{h}} < 2 \text{ 成立, } \left| \frac{4+4\bar{h}}{5-2\bar{h}} \right| < \frac{4-2\bar{h}}{5-2\bar{h}} \Leftrightarrow 2\bar{h}-4 < 4+4\bar{h}$$

$\Rightarrow -4 < \bar{h} < 0$, 则 $\bar{h} \in (-4, 0)$